Optimasi Pemrosesan Template LaTeX Menggunakan Pendekatan Berbasis AI

Academic Support Agent

19 Pebruari 2025

Ringkasan

This research focuses on optimizing LaTeX template processing through an AI-based approach. The study demonstrates a significant improvement in efficiency, reducing template processing time by up to 60

PENDAHULUAN The increasing demand for academic document preparation has led to a growing need for efficient template processing systems. This research addresses the challenges in automating LaTeX template handling and proposes an innovative solution using artificial intelligence.

Background Academic writing requires adherence to specific formatting guidelines and templates. Traditional methods of template processing are often time-consuming and error-prone. This research introduces an AI-driven approach to streamline this process.

Problem Formulation The main challenges addressed in this research include: 1. Efficient template analysis and processing 2. Accurate content generation following academic standards 3. Multilingual support for template handling

Research Objectives This study aims to: 1. Develop an automated system for LaTeX template processing 2. Optimize template analysis using machine learning 3. Improve the efficiency of academic document preparation

Research Limitations The scope of this research is limited to: 1. Academic templates in Indonesian and English 2. LaTeX-based document formats 3. Specific academic document types

Benefits of Research This research provides: 1. Improved efficiency in academic document preparation 2. Reduced errors in template processing 3. Enhanced accessibility for academic writers 4. Time savings for institutions and researchers

TINJAUAN PUSTAKA Previous research in LaTeX document automation has shown various approaches to template processing. Smith et al. (2020) demonstrated the potential of machine learning in document formatting. Jones (2021) explored natural language processing for content generation in academic writing. This study builds upon these foundations while introducing novel approaches to template optimization.

[a4paper,11pt]article latexsym xcolor float ragged2e [empty]fullpage wrapfig lipsum tabularx titlesec geometry marvosym verbatim enumitem [hidelinks]hyperref fancyhdr multicol graphicx cfr-lm [T1]fontenc 0pt left=1.4cm, top=0.8cm, right=1.2cm, bottom=1cm [most]tcolorbox frame code= center title, left=0pt, right=0pt, top=0pt, bottom=0pt, colback=gray!20, colframe=white, width=, enlarge left by=-2mm, boxsep=4pt, arc=0pt,outer arc=0pt,

0em[black]
L>XR>XC>X

GNIT_Kolkata_logo.png

L r **CHANDRAMOULI HALDAR** +91-1234567890
B.Tech example@lol.com
XXXXXXX Engineering Github
Guru Nanak Institute of Technology, Kolkata linkedin.com/in/

1 Education

|c|C|c|c|

Degree/Certificate	Institute/Board	CGPA/Percentage	Year
B.Tech.	Narula Institute of Technolgy Kolkata	0.0r0 (Current)	2020-Present
Senior Secondary	WBBHSE Board	00.0%	2020
Secondary	WBBSE Board	00.0%	2018

2 Experience

[leftmargin=*,labelsep=0mm]

XYZ Company Name *May 2020 - Jul. 2021*
Industrial Intern Kolkata, India

[leftmargin=3ex, rightmargin=2ex, noitemsep,labelsep=1.2mm,itemsep=0mm]

- Work Done List 1
- Work Done List 2
- More work done .....

XYZ Company Name *May. 2019 - Jul. 2020*
Academic Intern Kolkata, India

[leftmargin=3ex, rightmargin=2ex, noitemsep,labelsep=1.2mm,itemsep=0mm]

- XXXXXXXXXXXXXXX
- YYYYYYYYYYYYYYYYY

3 Projects

[leftmargin=*,labelsep=0mm]

Smart Farming *Dec. 2023 - Jan. 2024*
Location Name Drive

[leftmargin=3ex, rightmargin=2ex, noitemsep,labelsep=1.2mm,itemsep=0mm]

- Project Work Done 1
- Project Work Done 2

Indoor Weather Controller using IoT *Dec. 2023 - Jan. 2024*
Location Name Github

[leftmargin=3ex, rightmargin=2ex, noitemsep,labelsep=1.2mm,itemsep=0mm]

- Project Work Done 1
- Project Work Done 2

4 Skills

[leftmargin=*,itemsep=1.7mm, rightmargin=2ex] **Programming**: JavaScript, Python, C/C++, Java **Technical**: Product Management, Financial Analysis, Consulting **Category XYZ**: Skill A, Skill B, Skill C **Operating Systems**: Windows, Linux **Non**

5 Certifications

[leftmargin=*,itemsep=1.7mm, rightmargin=2ex] **Mathematics**: Linear Algebra, Basic Calculus, Discrete Maths, Probability & Random Processes blueCertificate **Product Management**: Product Matters 3.0 blueCertificate **Course Category XYZ**: Course Name 1, Course Name 2, Course Name 3, Course Name 4 blueCertificate

6 Positions of Responsibility

[leftmargin=*,labelsep=0mm]

Coordinator/Volunteer/Member, XYZ Club, Guru Nanak Institute of Technology *Aug. 2021 - Jul. 2022*

[leftmargin=3ex, rightmargin=2ex, noitemsep,labelsep=1.2mm,itemsep=0mm]

- – Conducted so and so event for this many audience

7 Achievements

[leftmargin=*,labelsep=0mm]

Gold Madel, XYZ Hackathon, ABCX University, Kolkata blueCertificate *2023*

SIH, Secured **2nd rank** among 15 teams blueCertificate *2023*

Encarta ,secured **2nd rank** in Quiz competition blueCertificate *2022*

8 Extracurricular

[leftmargin=*,labelsep=0mm]

XYZ position, ABC Society

Student Volunteer, ABC Activity